

3(a)	It is a <u>net upward force</u> acting on a body by a fluid when it is immersed in the fluid, and it occurs due to the <u>difference in pressure</u> between the <u>top and bottom surfaces</u> of the immersed portion of the body.	[1]
3(b)(i)	$M = \rho V$ $= \rho \times \frac{4}{3} \pi r^3$ $= 1.21 \times \frac{4}{3} \pi (11 \times 10^{-2})^3$ $= 0.00675 \text{ kg}$ $= 6.75 \text{ g}$	 [1] [1]
3(b)(ii)	Net force is zero. Thus, weight of balloon + weight of air in balloon = upthrust + normal contact force by scale on balloon $1.30 \times 10^{-3} g + m_{air} g = 6.75 \times 10^{-3} g + 1.55 \times 10^{-3} g$ $m_{air} = 7.00 \text{ g}$	 [1] [1]
3(b)(iii)	The air is being compressed by the elastic balloon material so is at a higher pressure and hence denser.	[1]
3(c)	As speed increases, drag force (acting in the upward direction) increases. Acceleration becomes zero when resultant force equals zero.	 [1] [1]

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No.	Solution	Remarks